

AfyaPlus Executive Evidence Brief

Decision: Approve controlled, quality-gated use of openai/gpt-4o-mini for the evaluated AfyaPlus workflows. Do not treat simulation results or this static brief as proof of production readiness or regulatory compliance.

Evidence boundary

- **Measured:** A live OpenRouter evaluation used the fixed AfyaPlus clinical dataset. openai/gpt-4o-mini completed **15/15 questions**, passed **7/7 clinical gates**, scored **5.0000/5 overall**, and achieved **100% grounding compliance** [E1, E2].
- **Measured:** openai/gpt-4o completed **15/15 questions** but achieved **93.33% grounding compliance**, so it failed the required **100%** gate and is not eligible for cost-led routing [E1, E2].
- **Simulated:** Month 2 first detected operational drift of **+563.582 ms latency** and **+103.408 tokens** per request. Month 3 added actionable quality drift: ROUGE-L **-0.110996**, overall score **-0.594667**, and grounding compliance **-20 percentage points** versus reference [E3].
- **Simulated:** The 30-day mixed-model workload contains **9,000 requests** and projects **USD 3.39612000 / KES 439.80**, consuming **94.34%** of the daily budget cap and producing a **WARNING** status [E4].
- **Live-only:** Current API health is intentionally not frozen into this report. The authenticated dashboard probes the real /health endpoint and reports UNKNOWN when the service cannot be reached [E6].

Leadership recommendation

- Put safety before price: retain the all-gates-pass requirement and route only to models that satisfy it.
- Prefer openai/gpt-4o-mini for this evaluated scope. Simulated all-mini routing preserves the recorded gates and projects **USD 0.72216000** over 30 days, saving **USD 2.67396000 (78.74%)** against the mixed baseline [E5].
- Treat any actionable quality-drift alert as a release hold: investigate, re-evaluate, and restore gate compliance before expansion.
- Use the authenticated dashboard for live health, aggregate API metrics, drift, quality, and budget decisions. Keep Prometheus network-restricted.
- Require human clinical and governance review before broader deployment; these results do not establish legal compliance, patient safety in every scenario, or end-to-end production readiness.

Source register

- [E1] evaluation/model_comparison_summary.csv — columns model, total_questions, successful_questions, overall, all_gates_passed.
- [E2] evaluation/quality_gate_log.csv — columns model, gate, threshold, observed, passed.
- [E3] drift/drift_trend_table.csv — columns month, column, mean_change, drift_detected, requires_action.
- [E4] cost/cost_projection_30d.csv — overall-row columns projected_requests, projected_cost_usd, projected_cost_kes, daily_budget_usd, budget_utilization, budget_status.
- [E5] cost/structural_savings_analysis.csv — columns status, selected_model, quality_preserved, proposed_30d_usd, savings_usd, savings_percent.
- [E6] dashboard/health.py — probe_health; dashboard/README.md — “Architecture and failure boundaries”.